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Team

March Metrics

Brandeis Datathon 2026

Predicting March Madness Through Data & Analytics



Executive Summary

01.

Higher seeds (1–4) dominate, but do not fully determine outcomes

02.

Upsets occur in ~28% of games (mid-seed matchups)

03.

Efficiency and talent outperform seeding

04.

Random Forest model: 76% accuracy

05.

Final Four: Tennessee, St. John's, Michigan State, Arizona

06.

Winner: Tennessee

Problem & Approach

Problem

Predict tournament outcomes
+ identify success drivers

Approach

Cleaned data and removed
leakage

Problem

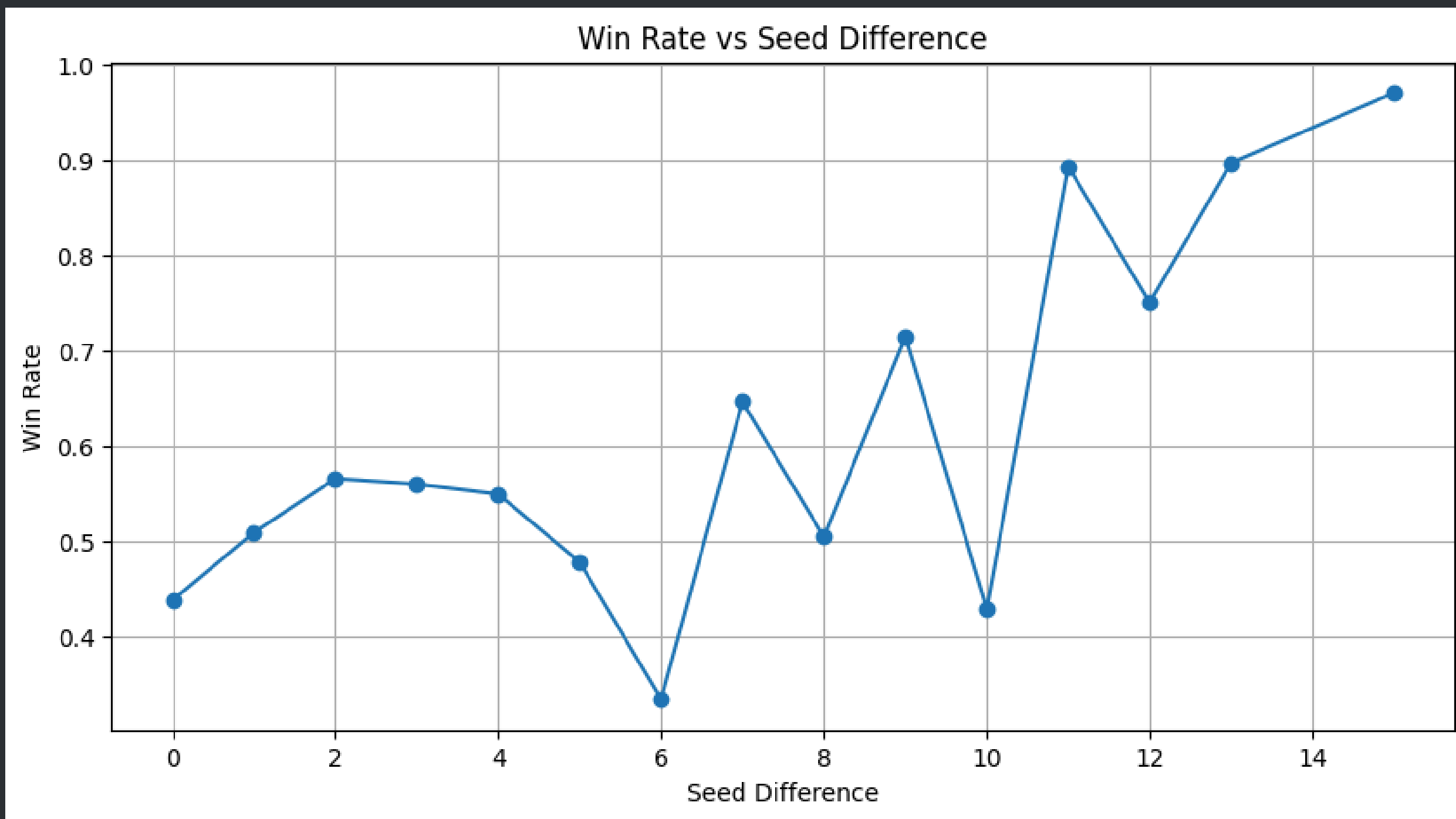
Multiple datasets (team stats,
performance metrics)

Approach

Created matchup-level difference
features (Team A – Team B)



Seed Performance

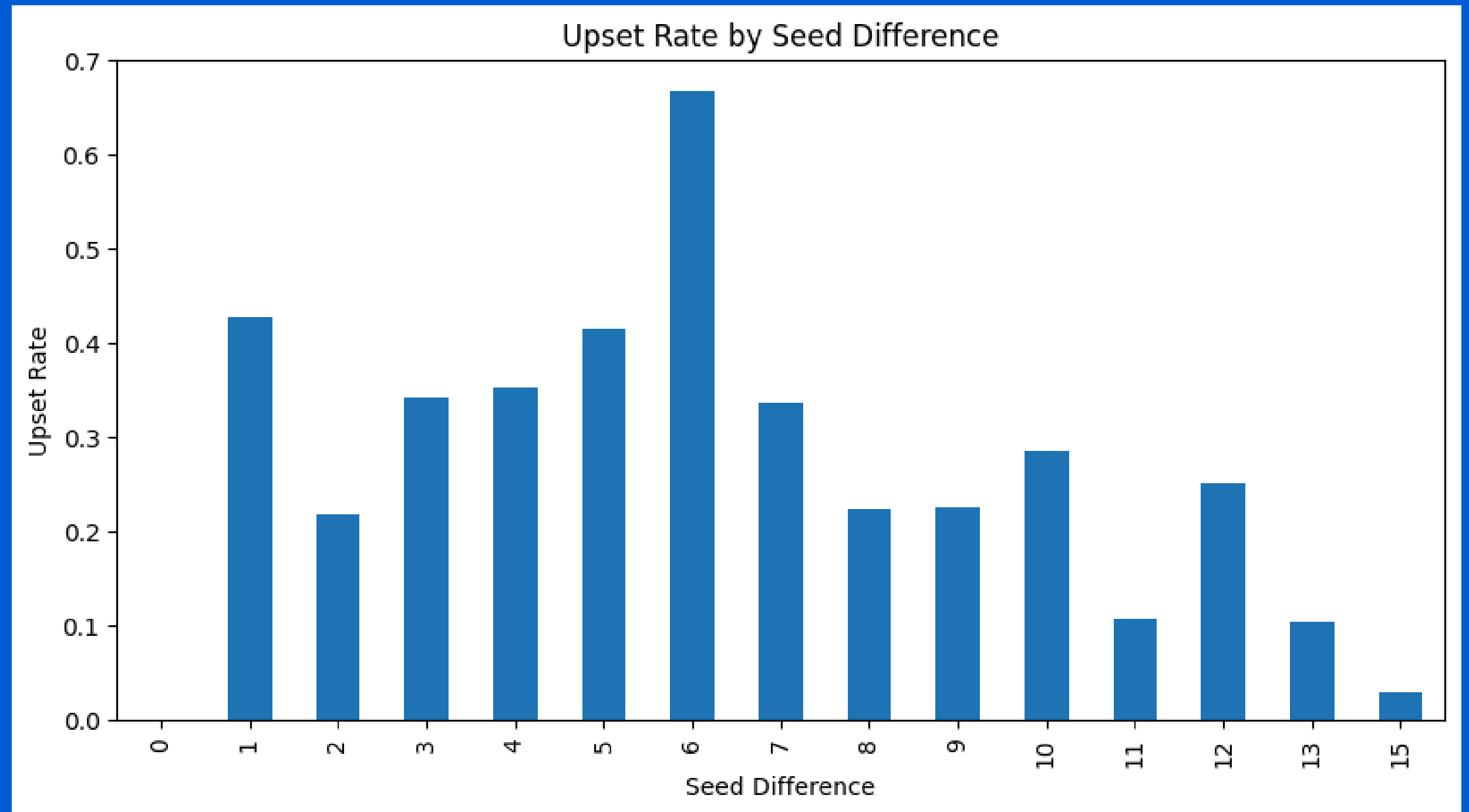


- Higher seeds consistently achieve higher win rates
- Seeds 1–4 dominate early tournament rounds
- Lower seeds (10–16) rarely advance
- Seed success is quantified using win rates and upset frequency

Upsets & Tournament Unpredictability

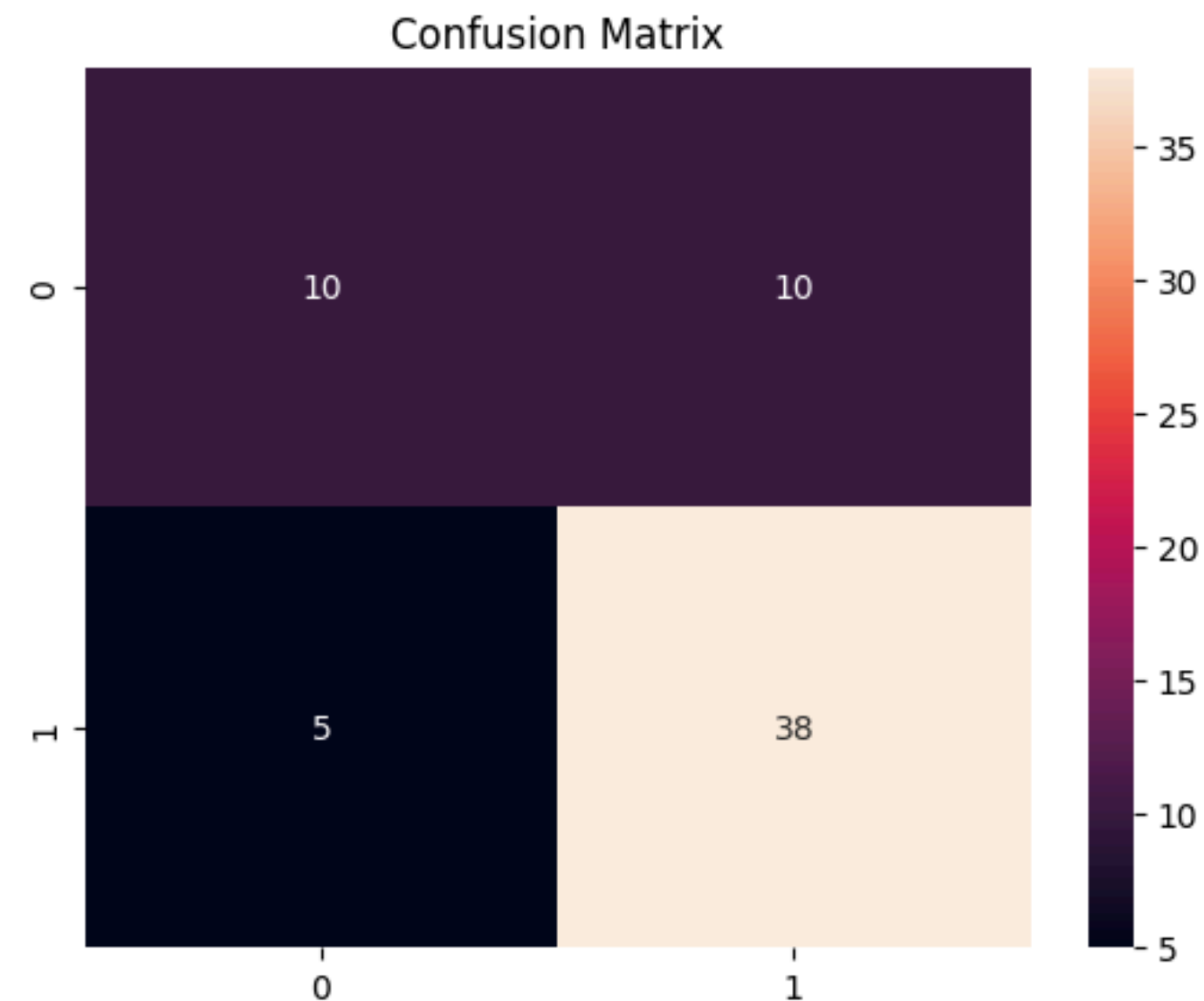


- Upsets occur in ~28% of tournament games
- Most upsets occur in mid-seed matchups (5 vs 12, 6 vs 11)
- Small seed differences lead to high unpredictability
- Large seed gaps result in more predictable outcomes



Model & Performance

- Built predictive models using matchup-level features
- Logistic Regression Accuracy: 71%
Random Forest Accuracy: 76% (Best Model)
- Random Forest captures complex, non-linear relationships
- Model trained on historical tournament data (tested on latest year)



Key drivers:

- Offensive efficiency
- Defensive strength
- Talent

2026 Tournament Predictions

Final Four

1. Tennessee
2. St. John's
3. Michigan State
4. Arizona



Teams ranked using model-predicted win probabilities.

Predictions based on:

- Offensive efficiency
- Defensive efficiency
- Team talent

“Teams were selected based on highest predicted win probabilities and strong efficiency + talent metrics.”

Conclusion

Efficiency & Talent > Seeding

Upsets (~28%) drive unpredictability

Random Forest: 76% Accuracy

Predictions are probabilistic

Data → smarter decisions

“From bracket chaos to data clarity - decoding March Madness with analytics.”

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